

0 8

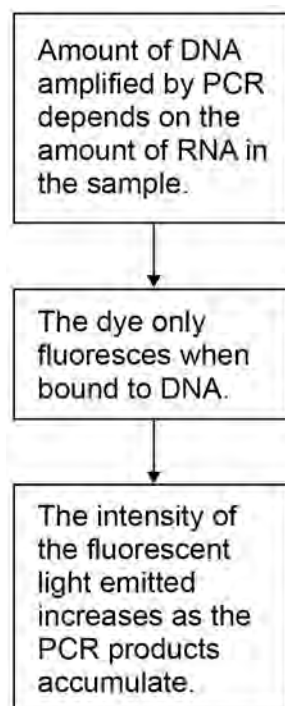
One way to detect and measure accurately the amount of RNA in a tissue sample is by RT-PCR (reverse transcriptase-polymerase chain reaction).

RT-PCR uses a reaction mixture containing:

- the sample for testing
- reverse transcriptase
- DNA nucleotides
- primers
- DNA polymerase
- fluorescent dye.

The principle behind this method is shown in **Figure 8**.

Figure 8



0 8 . 1

Explain the role of reverse transcriptase in RT-PCR.

[1 mark]

0 8 . 2

Explain the role of DNA polymerase in RT-PCR.

[1 mark]

0 8 . 3

Any DNA in the sample is hydrolysed by enzymes before the sample is added to the reaction mixture.

Explain why.

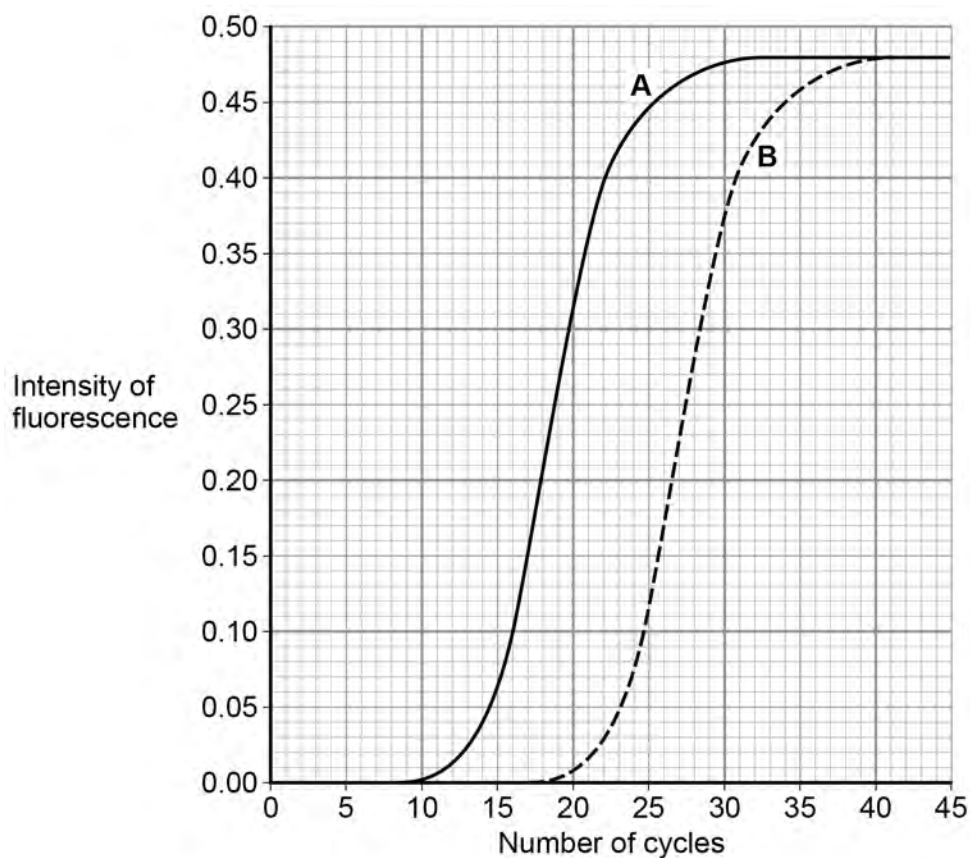
[2 marks]

Question 8 continues on the next page

0 8 . 4

Figure 9 shows the results from using RT-PCR to detect RNA in two different samples, **A** and **B**.

Figure 9



A quantitative comparison can be made of the amount of RNA in samples **A** and **B**. This involves determining the number of cycles required to reach 50% maximum concentration of DNA (**C**).

The amount of RNA in a sample can be measured as: $\frac{1}{C}$

Use this information to calculate the ratio for RNA content in sample **A** : RNA content in sample **B**.

[2 marks]

Answer = _____

0 8 . 5

Suggest **one** reason why DNA replication stops in the polymerase chain reaction.**[1 mark]**

0 8 . 6

Scientists have used the RT-PCR method to detect the presence of different RNA viruses in patients suffering from respiratory diseases.

The scientists produced a variety of primers for this procedure.

Explain why.

[2 marks]

Turn over for the next question**Turn over ►**

